

Word Meaning and Semantic Relations

Francis **Bond**

Department of Asian Studies,
Palacký University, Olomouc, Czechia

[<bond@ieee.org>](mailto:bond@ieee.org)

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Roadmap

1 The meanings of words

2 Lexical Semantics

3 Wordnet

4 Lexical Universals



Words carry different meanings: **leave** I

- 10070 *Nothing was left save a few acres of ground , and the two-hundred-year-old house , which is itself crushed under a heavy mortgage .*
- 10079 *The money which my mother had left was enough for all our wants , and there seemed to be no obstacle to our happiness . ”*
- 10085 *He had no friends at all save the wandering gipsies , and he would give these vagabonds leave to encamp upon the few acres of bramble- covered land which represent the family estate , and would accept in return the hospitality of their tents , wandering away with them sometimes for weeks on end .*
- 10107 *She left her room , therefore , and came into mine , where she sat for some time , chatting about her approaching wedding .*



Words carry different meanings: *leave* II

- 10108 *At eleven o'clock she rose to leave me , but she paused at the door and looked back.*
- 10439 *" The rest you will leave in our hands . "*
- 10449 *And now , Miss Stoner , we must leave you for if Dr. Roylott returned and saw us our journey would be in vain .*
- 10526 *Then he turned down the lamp , and we were left in darkness .*

How many different meanings?

?

From the **NTU Multilingual Corpus** (*Adventure of the Speckled band*, concept lemma = *leave*)



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How can we represent the differences?

- Definitions
- Translations/paraphrases
- Semantic Relations
- Components
- Word Embeddings



Semantic Representations of Words

- Divide meaning into
 - ▶ **reference**: the relation to the world/mental space
 - ▶ **sense**: the rest of the meaning
 - **denotation** the part that distinguishes the meaning from other meanings
 - **connotation** cultural or emotional associations
- Introduce **concepts**
 - ▶ How can we represent concepts?
 - ▶ How do we learn them?
 - Typically children start off by **underextending** or **overextending** concepts
- Example: *That dog*
 - ▶ reference — the animal over there
 - ▶ sense — canine quadruped domesticated by man
 - ▶ connotation — faithful, friendly (or dirty)



Definitional Semantics

- Standard lexicographic approach to lexical semantics:
semantics = *the study of language meaning*
tailor = *a person whose occupation is making and altering garments*
- Definitions are conventionally made up of;
 - ▶ **genus**: what class the lexical item belongs to
 - ▶ **differentiae**: what attributes distinguish it from other members of that class
- Often hard to understand if you don't already know the meaning!



Definitional Semantics: pros and cons

- Pros:
 - ▶ familiarity (we are taught to use dictionaries)
- Cons:
 - ▶ subjectivity in sense granularity (splitters vs. lumpers) and definition specificity
 - ▶ circularity in definitions
 - ▶ consistency, reproducibility, ...
 - ▶ often focus on diachronic (historical) rather than synchronic (current) semantics



Entries for *leave* I

- 02015598-v (72) V1, V2 *get out, go out, leave, exit* “move out of or depart from”
- 02356230-v (8) V3 *leave, entrust* “put into the care or protection of someone”
- 02009433-v (149) V1 *leave, go away, go forth* “go away from a place”
- 02229055-v (7) V3 *leave, will, bequeath* “leave or give by will after one’s death”
- 02729414-v (56) V2 *leave* “act or be so as to become in a specified state”
- 02730135-v (5) V2 *leave* “have left or have as a remainder”
- 06690114-n (1) *leave* “permission to do something”
- Not to be confused with left hand and the leaves fell,



Paraphrases and translation

- Saying the same thing in different words
 - ▶ Same language = **paraphrase**
 - ▶ Different language = **translation**
- We showed some paraphrases in the entries given above
- If you speak another language, then you can use that to disambiguate many things.
 - ▶ **leave, entrust [something, to someone]** = 預ける *azukeru*
 - ▶ **get out, go out, leave, exit [somewhere]** = 去る *saru*
 - ▶ **leave, will, bequeath [something, to someone]** = 遺す *nokosu*
 - ▶ **leave [something, somewhere]** = 残す *nokosu*
 - ▶ **remain** = 残る *nokoru*

Can you explain the ambiguity in *The money which my mother had left was enough for all our wants?*



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- Lexical semantics is concerned with the identification and representation of the semantics of lexical items
- But a given word can have multiple interpretations
 - ▶ **Polysemy**: having multiple meanings
 - ▶ **Monosemy**: having only one meaning
- **Homonyms** are words with two unrelated meanings:
 - ▶ **homographs**: same spelling
bow vs *bow*; *keep* vs *keep*
 - ▶ **homophones**: same pronunciation
right vs *write*; *keep* vs *keep*

Distinguishing Polysemes

- **Antagonism**: can the word be used in a sentence with multiple competing interpretations that are incompatible?

Kim can't bear children

- ▶ Cannot have children
- ▶ Doesn't like children

- **Zeugma**: can the word be used in a context where multiple competing interpretations are simultaneously evoked?

Kim and her visa expired

- ▶ died
- ▶ ran out

Hitmen were quite expensive, so she decided to take out a loan and her husband.

?

- **Paraphrase/Translation**: Is there more than one (clearly different) way to paraphrase/translate the word.



Necessary and Sufficient Conditions

- Can we define words in terms of **conditions**?
 - ▶ **zebra**
 - quadruped
 - animal **redundant**
 - black and white striped
 - herbivore
- These are **intrinsic**, **generic** properties
 - ▶ An albino zebra with three legs is still a zebra
- Can we use words even if we don't know their properties?
 - ▶ **Kway Teow**
- We seem to be ok with fairly vague definitions
 - ▶ What is a **dog-cart**?
 - ▶ What is a **grass snake**?
 - ▶ What is a **swamp adder**?



Words/Concepts are related in many ways

We can also look at words (or more properly senses) in terms of their relations to other words.

- **Hyponymy/Hypernymy**
- **Synonymy**
- **Antonymy** (Opposites)
- **Meronymy**
 - ▶ **Member-Collection**
 - ▶ **Portion-Mass**
 - ▶ **Element-Substance**
- **Domain** (lexical field)



Hypernymy and Hyponymy

- **Hyponymy**: X is a hyponym of Y iff $f(X)$ entails $f(Y)$ but $f(Y)$ does not entail $f(X)$ (for all or most f):

Kim has a pet dog $\models$ Kim has a pet animal

Kim has a pet animal $\not\models$ Kim has a pet dog

N.B. complications with universal quantifiers and negation:

Kim likes all animals $\models$ Kim likes all dogs

Kim likes all dogs $\not\models$ Kim likes all animals

- **Hypernymy**: Y is a hypernym of X iff X is a hyponym of Y
- Can a word have multiple hypernyms?
 - (1) *tank₁ $\subset$ military_vehicle₁; $\subset$ tracked_vehicle₁; $\subset$ armored_vehicle₁; ? $\subset$ weapon₁*



What is **entailment**

Entailment ($\models$): *A sentence p entails a sentence q when the truth of the first (p) guarantees the truth of the second (q), and the falsity of the second (q) guarantees the falsity of the first (p).*



Properties of hypernymy/hyponymy

- Asymmetric; applies at the sense level
- applies only to lexical items of the same word class
- Transitive: *dog*₁ $\subset$ *mammal*₁ $\subset$ *animal*₁



Not everything is lexicalized

Lexicalization is the process by which new words, having gained widespread usage, enter the lexicon.

neutral (Hyper)	male	female	child
<i>sheep</i>	<i>ram</i>	<i>ewe</i>	<i>lamb</i>
<i>cow</i>	<i>bull</i>	<i>cow</i>	<i>calf</i>
<i>goose</i>	<i>gander</i>	<i>goose</i>	<i>gosling</i>
<i>horse</i>	<i>stallion</i>	<i>mare</i>	<i>foal:colt/filly</i>
<i>dog</i>	<i>dog</i>	<i>bitch</i>	<i>puppy</i>
<i>snake</i>			

snake is not ambiguous it is **vague**: the meaning is underspecified.

Can you do this for *pig*, *cat* or *chicken*?

Can you give an example of this in another language?

Language Change and Auto-hyponyms I

- The meanings of words change over time
 - ▶ **guitar** — “a stringed instrument usually having six strings”: originally these all used the body to make sound
 - ▶ We then get **electric guitar** — “a guitar with a built-in pickup or pickups which convert string vibrations into electrical signals for amplification”
 - ▶ To refer to non-electric guitars we get a new coining **acoustic guitar** – “a guitar that does not require electrical amplification”: which used to just be guitar
- **guitar** is now a hypernym of them both and can refer to either
- we can also refer to the prototypical guitar (acoustic) using reduplication

What kind of guitar do you play? Guitar guitar



Language Change and Auto-hyponyms II

- Sometimes this practice becomes politically charged, although linguistically it is unremarkable
 - ▶ **woman** “an adult female person”
 - ▶ **trans woman** “a person who identifies as a woman but was assigned male at birth”
 - ▶ **cis woman** “a person who identifies as a woman and was assigned female at birth”

Can you give other examples of this in English or other languages?



- **Propositional synonymy**: X is a propositional synonym of Y if
 - ▶ (i) X and Y are syntactically identical,
 - ▶ (ii) substitution of Y for X in a declarative sentence doesn't change its truth conditions

e.g., *violin* and *fiddle*

- Why propositional synonymy is over-restrictive:
 - ▶ syntactic identity (cf. *eat* and *devour*)
 - ▶ collocations (cf. *cemetery* and *graveyard*)
 - ▶ gradability (cf. *sofa/settee* vs. *boundary/frontier*)

Near Synonymy

- Near synonyms are substitutable in **some/most** rather than **all** contexts
- Synonymy via semantics: synonyms share “common traits” or attributional overlap, walking the fine line between “necessary resemblances” and “permissible differences”:

grain vs. *granule*; *green* vs. *purple*; *alsation* vs. *spaniel*

- Permissible differentiation via **clarification**.

Here is a grain, or granule, of the substance.

** The cover is green, {or, that is to say} purple.*

** He likes alsations, in other words, spaniels*



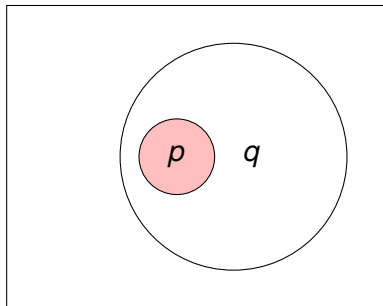
Properties of synonymy

- Symmetric
- traditionally applies only to lexical items of the same word class but pairs like *can* vs *be able to* suggest otherwise
- applied at the sense level
- $\approx$ converse of polysemy

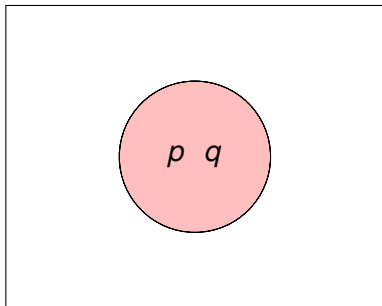


Semantic Relations as Sets ($p \subset q$ and $p \sim q$)

$p \subset q$ **hypernym**



$p \sim q$ **synonym**



Antonymy (opposites) I

- **Simple antonyms:** the negative of one implies the positive of the other.
 - (2) *dead/alive*
 - (3) *pass/fail*
- **Gradable Antonyms:** points along a scale
 - (4) *boiling/hot/warm/tepid/cool/cold/freezing*
 - (5) *fascinating/interesting/dull/boring*
- **Reverses:** reverse the direction of a motion
 - (6) *ascend/descend*
 - (7) *up/down; right/left*



Antonymy (opposites) II

- **Converses**: the same act from different points of view

(8) *above/below; right/left*

(9) *employer/employee*

(Slightly non-standard usage by Saeed)

- **Taxonomic Sisters**: children of the same (grand)parent

(10) *Monday/Tuesday/.../Sunday*

in WordNet: *day of the week* $\supset$ *weekday*, *weekend*

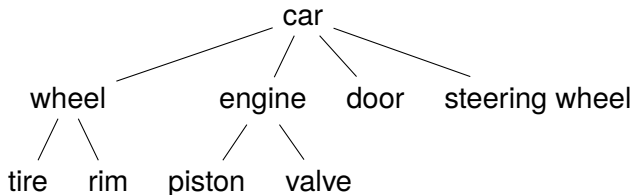
(11) *LMS/English/Chinese/...*

Context dependent

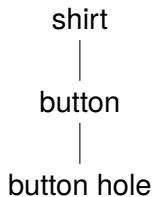


Meronymy

- **Meronymy** refers to the part-whole relation
 - ▶ **meronym** is the part
 - ▶ **holonym** is the whole



- It is not always transitive:



We don't normally say that a **button hole** is part of a **shirt**.



Member-Collection

- The relation between a collection and one of the units that makes it up
 - (12) *tree–forest*
 - (13) *sheep–flock*
 - (14) *fish–school*
 - (15) *book–library*
 - (16) *member–band*
 - (17) *musician–orchestra*
 - (18) *student–class*



Portion-Mass

- The relation between a mass noun and a typical unit of measurement

(19) *drop–liquid*

(20) *grain–sand/salt/truth*

(21) *sheet/ream–paper*

(22) *lump–coal (or just about anything)*

(23) *strand–hair*

(24) *rasher–bacon*

- Similar to classifiers in many ways, e.g. in Malay

(25) *ekor* “tail”–*animal*

(26) *orang* “human”–*person*



Domain (lexical field)

The domain in which a word is typically used with this meaning.

- (27) **driver**₁ — the operator of a motor vehicle
- (28) **driver**₂ — someone who drives animals that pull a vehicle
- (29) **driver**₃ — a golfer who hits the golf ball with a driver [**golf**]
- (30) **driver**₄ — ($\simeq$ device driver) a program that determines how a computer will communicate with a peripheral device [**computer science**]
- (31) **driver**₅ — ($\simeq$ number one wood) a golf club (a wood) with a near vertical face that is used for hitting long shots from the tee [**golf**]

Some **golf** terms: **approach**₉, **approach shot**₁, **golf course**₁, **links course**₁, **wedge**₅, **tee**₁, **scratch**₉, **putt**₁, **slice**₁, **hook**₁



And More

- There are many, many more lexical relations advocated by various theories including:
 - ▶ Troponymy/hypernymy (cf. *walk* vs. *lollop*) “way of doing something”
 - ▶ Entailment (cf. *snore* vs. *sleep*) “if you do one thing, you must be doing the other”
 - ▶ Operator (cf. *question* vs. *ask*) “the thing you do by doing something”
 - ▶ Magnifier (cf. *wound* vs. *badly*) “intensifier, diminisher”
 - ▶ Usage (cf. *strong-willed* vs. *pig-headed* “stubborn”)
pig-headed is **pejorative**



Derivational Relations

- Often words are linked by more or less systematic relations, sometimes morphologically marked
 - ▶ *beauty/beautiful/beautify*
 - ▶ *cute/cuteness*
- These can be very productive, but can also change the meaning of the derived word



Agentive Nouns

- An **agentive noun** is a word that is typically derived from another word denoting an action, and that identifies an entity that does that action.

verb + **-er, -or, -ant**

(32) *murderer, commentator, whaler, director, computer*

(33) ?? *undertaker, cooker, footballer* (Saeed also includes these)

- Should *murderer* be listed separately from *murder* in the dictionary? Why or why not?
- Also **recipient nouns** that show the undergoer: **verb** + **-ee**:
employee, trustee



Agentive Nouns in Other Languages I

- Slovak

- ▶ **-ník/-níčka**: more concrete or traditional

(34) **pracovať** (to work) - **pracovník** (m.), **pracovníčka** (f.)

(35) **účtovať** (to do accounting) - **účtovník** (m.),
účtovníčka (f.)

(36) **poľovať** (to hunt) - **poľovník** (m.), **poľovníčka** (f.)

- ▶ **-teľ/-telka**

(37) **učiť** (to teach) - **učiteľ** (m.), **učiteľka** (f.)

(38) **organizovať** (to manage) - **organizovateľ** (m.),
organizovateľka (f.)

(39) **riadiť** (to lead) - **riaditeľ** (m.), **riaditeľka** (f.)

*the word riaditeľ does not mean a leader but it is used in a school for headmaster



Agentive Nouns in Other Languages II

- ▶ **-ár/árka**: this suffix is used for someone who works with/produces/deals with something

(40) **piecť** (to bake) - **pekár** (m.), **pekárka** (f.)

(41) **ryba** (fish) - **rybár** (m.), **rybárka** (f.)

(42) **víno** (wine) - **vinár** (m.), **vinárka** (f.)

- ▶ **-ista/-istka** this suffix is used for borrowed/lent words; an "international suffix"

(43) **futbal** (noun: football) - **futbalista** (m.), **futbalistka** (f.)

(44) **basketbal** (noun: basketball) - **basketbalista** (m.), **basketbalistka** (f.)

(45) **volejbal** (noun: volleyball) - **volejbalista** (m.), **volejbalistka** (f.)

Thanks to Emma Martincova for the Slovak examples



Agentive Nouns in Other Languages III

- Japanese (suffix distinguishes person/machine)
 - ▶ 運転する → 運転者 *untēn-sha* “driver”
 - ▶ 計算する → 計算者 計算機 *keisan-sha/ki* “computer”
 - ▶ 研究する → 研究者 研究員 *kenkyū-sha/in* “researcher”
 - ▶ 読む → 読み手 読者 *yomite/dokusha* “reader”
- Malay (prefix can convert any part of speech)
 - ▶ *bantu* (v) “help” → *pembantu* “assistant/helper”
 - ▶ *potong* (v) “cut” → *pemotong* “cutter (human/machine)”
 - ▶ *terbang* (v) “fly” → *penerbang* “pilot (not passenger)”

Thanks to Yeo Jia Qi for the Malay examples



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1 The meanings of words

2 Lexical Semantics

3 **Wordnet**

4 Lexical Universals



- WordNet is an open-source electronic lexical database of English, developed at Princeton University
<http://wordnet.princeton.edu/>
- Made up of four separate semantic nets, for each of nouns, verbs, adjectives and adverbs
- WordNets exist for many languages, my group has worked on:
 - ▶ Japanese
 - ▶ Bahasa Malay/Indonesian
 - ▶ Chinese (Mandarin and Cantonese)
 - ▶ The shared open multi-lingual wordnet (150+ languages)
<https://omwn.org/>
 - ▶ Kristang
 - ▶ Myanmar
 - ▶ Czech



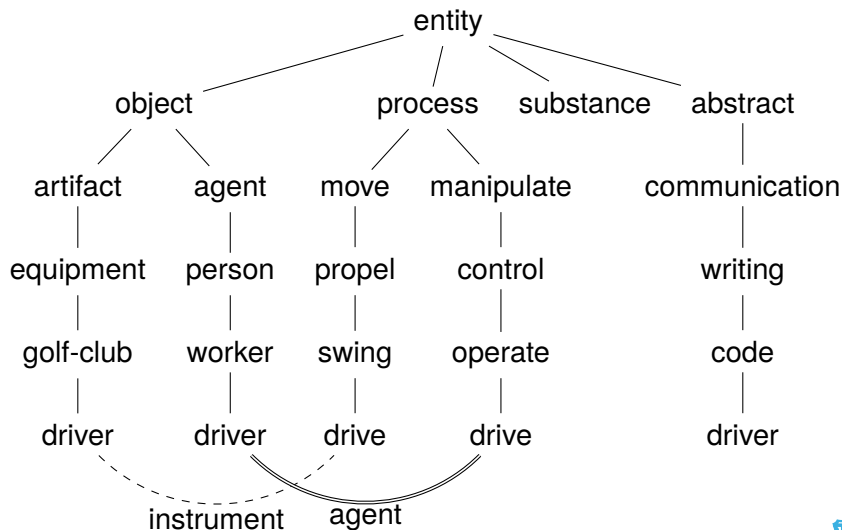
- Lexical items are categorised into $\sim 115\text{K}$ (and counting) glossed **synsets** (= synonym sets)
- Lexical relations at either the synset level or sense (= combination of lexical item and synset) level
- Strongly lexicalist (originally):
 - ▶ synsets only where words exist
 - ▶ but many multiword expressions ($\approx 50\%$)

Psycholinguistic Foundations of WordNet

- Strong foundation on hypo/hypernymy (lexical inheritance) based on
 - ▶ response times to sentences such as:
a canary {can sing/fly, has skin}
a bird {can sing/fly, has skin}
an animal {can sing/fly, has skin}
 - ▶ analysis of anaphora:
I gave Kim a novel but the {book, ?product,...} bored her
Kim got a new car. It has shiny {wheels, ?wheel nuts,...}
 - ▶ selectional restrictions
- Is now often used to calculate **semantic similarity**
 - ▶ The shorter the path between two synsets the more similar they are
 - ▶ Or the shorter the path to the nearest shared hypernym, ...



Word Meaning as a Graph



Wordnet in this course

- We will use wordnet as a source of examples and inspiration
- There is a good online interface here:
<https://cygnet.maudslay.eu/>
- My students have used wordnets for:
 - ▶ Japanese derivational relations (Bond and Wei, 2019)
 - ▶ pronoun representation for Japanese, Mandarin and English (Seah and Bond, 2014)
 - ▶ exclamatives and classifiers (Mok et al., 2012; Morgado da Costa and Bond, 2016)
 - ▶ sentiment analysis (Le et al., 2016; Bond et al., 2019)
 - ▶ cross-lingual sense annotation (Bonansinga and Bond, 2016)
 - ▶ multilingual crosswords (Tan, 2012)
- ...
- If you are interested in doing something with it, come and talk to me!



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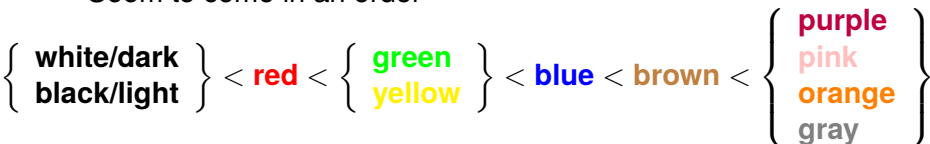


Color Terms

- **Basic Color Terms**

- ▶ Monolexemic
- ▶ Not a hyponym of any other color
- ▶ Can be widely applied
- ▶ Not derived from a noun

- **Focal Colors** are related to the neurophysiology of our visual system
- Seem to come in an order



Note: **pink** is really light **red**, and **orange** is light **brown**, ...

See also: <http://wals.info/chapter/133>: Colour Terms by Paul Kay and Luisa Maffi and (Kay and Maffi, 2000)



Core Vocabulary I

- Some universal terms can be used to compare languages
 - ▶ lexicostatistics (quantitative language relatedness assessment)
 - ▶ glottochronology (language divergence dating)
- The **Swadesh list**, developed by Morris Swadesh from 1940 onward
- Chosen for their universal, culturally independent availability in as many languages as possible



Core Vocabulary II

I, You, we, this, that, who, what, not, all, many, one, two, big, long, small, woman, man, person, fish, bird, dog, louse, tree, seed, leaf, root, bark, skin, flesh, blood, bone, grease, egg, horn, tail, feather, hair, head, ear, eye, nose, mouth, tooth, tongue, claw, foot, knee, hand, belly, neck, breasts, heart, liver, drink, eat, bite, see, hear, know, sleep, die, kill, swim, fly, walk, come, lie, sit, stand, give, say, sun, moon, star, water, rain, stone, sand, earth, cloud, smoke, fire, ash(es), burn, path, mountain, red, green, yellow, white, black, night, hot, cold, full, new, good, round, dry, name

- Available in many languages (hundreds);
- Now linked to wordnet (**Morgado da Costa et al., 2016**)



Conclusions

- We learned a little about word meaning, semantic relations, wordnet and universals
- This is covered in much more detail in **Saeed (2009, Chapter 3)** and **Kroeger (2022, Chapter 6 and 7)**



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Glossary of Key Terms (English–Czech) I

English	Česky
agentive noun	agentní podstatné jméno
antagonism	antagonismus
antonymy	antonymie (včetně stupňovatelnosti)
conditions	podmínky
connotation	konotace
contrasting	kontrastní
converses	opak
denotation	denotace
domain	doména
element–substance	prvek–látka
entailment	implikace
generic	obecný
holonymy	holonymie
homographs	homografy



Glossary of Key Terms (English–Czech) II

homonyms	homonyma
homophones	homofony
hypernymy	hyperonymie
hyponymy	hyponymie
incompatibility	nekompatibilita
intrinsic	vnitřní / vlastní
member–collection	člen–kolekce
meronymy	meronymie
monosemy	monosémie
overextending	nadměrné rozšíření významu
underextending	nedostatečné rozšíření významu
paraphrase	parafráze
pejorative	pejorativní
polysemy	polysémie
portion–mass	část–množství
primitives	prvky / primitiva



Glossary of Key Terms (English–Czech) III

propositional synonymy	propozicionální synonymie
recipient noun	recipientní podstatné jméno
reference	reference
relevant	relevantní
reverse	reverzní
semantic components	sémantické komponenty
semantic similarity	sémantická podobnost
sense	význam
synonymy	synonymie
taxonomic sisters	taxonomické sestry
translation	překlad
zeugma	zeugma

